

Diagnosable Evaluation of General Biswapped Networks *

Jheng-Cheng Chen and Chang-Hsiung Tsai[†]

Department of Computer Science and Information Engineering
National Dong Hwa University
Hualien, Taiwan 970, R.O.C.

Abstract

General biswapped networks were introduced in [4] as interconnection networks amenable to implementation using a mix of electronic and optical communication links. The model is inspired by and extended from Biswapped Networks and OTIS networks. In this paper, we focus on the diagnosabilities of general biswapped networks under the precise and pessimistic strategies based on the PMC diagnostic model. It is proved that the general biswapped graphs $Gbsw(G, H)$ with $\delta(G) \geq 1$ and $\delta(H) \geq 1$ is $\delta(Gbsw(G, H))$ -diagnosable and t_1/t_1 -diagnosable under the precise and the pessimistic strategies, respectively, where $t_1 = \min_{u, v \in V} \{|N(\{u, v\})|\}$.

1 Introduction

Multiprocessor systems are vulnerable to faults, and fault-tolerant computing has become very important for systems with several processors linked by an interconnection network. To maintain the reliability of a system, any faulty processor should be replaced with a fault-free one. The method of finding faulty processors is called fault diagnosis of the system. The degree of diagnosability of the system denotes the maximum number of faulty processors that can be accurately identified in the system.

System-level diagnosis is an important technique for improving system reliability and availability. A graph-theoretical model of system-level fault diagnosis termed the PMC model was proposed in a classic paper by Preparata, Metze and Chien [11]. In this model, every processor in a system performs tests on its neighbors utilizing communication links between them. When one pro-

cessor tests another, the tester declares the tested processor to be fault-free or faulty depending on the test response; the result is always accurate if the tester is fault-free, but if the tester is faulty, the result is unreliable. According to the traditional *precise diagnosis* strategy, it is demanded that all processors be identified correctly, specifically that all fault-free processors are identified as ‘fault-free’ and all faulty processors are identified as ‘faulty’ [11]. The pessimistic diagnosis strategy proposed by Kavianpour and Friedman [5] is a process to diagnose faults that allows all faulty processors can be isolated within a set having at most one fault-free processor. In order to identify faults, a number of tests must be performed among processors and the collection of all test results is referred to as a *syndrome*.

Under the PMC model with a pessimistic strategy, the diagnosabilities of some well-known multicomputer systems for example hypercubes [6], enhanced hypercubes [12], Möbius cubes [3], and star graphs [7] have been determined in the last few decade. Recently, Chang et al. [1] investigated the pessimistic diagnosabilities of regular networks and provided sufficient conditions for computing the pessimistic diagnosabilities of regular networks.

General biswapped networks were introduced in [4] as interconnection networks amenable to implementation using a mix of electronic and optical communication links. The model is inspired by and extended from Biswapped Networks and OTIS networks [10, 13, 14, 15]. A general biswapped network (GBSN) $Gbsw(G, H)$, where G is the base graph and H is the network graph, is composed of m copies of some n -node basis network G and n copies of some m -node basis network H . Such a network uses a simple rule for connectivity that ensures its semi-regularity, modularity, fault tolerance, and algorithmic efficiency. In particular, the BSNs are special cases in which the two basis networks are the same. The Ex-

*This work was supported in part by the National Science Council of the Republic of China under Contract NSC 100-2115-M-259-008-.

[†]Correspondence to: e-mail: chtsai@mail.ndhu.edu.tw.

changed Hypercube is another example of GBSNs, i.e., $EH(s, t)$ is isomorphic to $Gbsw(H(s), H(t))$.

The objective of this paper is to study the diagnosability of general biswapped networks under the two above-mentioned diagnosis strategies. It is proved that a $Gbsw(H, G)$ is t -diagnosable and t_1/t_1 -diagnosable under the precise and pessimistic strategies, respectively, where t is its degree and $t_1 = \min_{u,v \in V} \{|N(\{u, v\})|\}$.

The rest of this paper is organized as follows. Section 2 gives some definitions and notations. Then, the diagnosability of general biswapped networks under the precise and pessimistic diagnosis strategies are studied in Section 3. Finally, we conclude the paper in Section 4.

2 Definitions and Basic Properties

The topology of a multicomputer system is often represented by a graph $G = (V, E)$, where each of the vertices $u \in V$ denotes a processor and each edge $(u, v) \in E$ denotes a bidirectional communication link between two processors u and v . Hereafter, we speak of vertex and edge instead of processor and communication link. Throughout this paper the three terms - multicomputer, system and graph - will be used interchangeably. In this paper, we focus on an undirected graph without loops and follow [9] for the graph definitions and notations. $G = (V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) \mid (u, v) \text{ is an unordered pair of } V\}$. We say that V is the *vertex set* and E is the *edge set*. Two vertices u and v are *adjacent* if $(u, v) \in E$; the vertex u is called a neighbor of v , and vice versa. The degree of vertex u is denoted by $deg(u)$ and the degree of graph G is denoted by $\delta(G) = \min_{u \in V(G)} deg(u)$. A graph G' is a subgraph of a graph G if $V(G') \subseteq V(G)$ and $E(G') \subseteq E(G)$.

He et al. [4] proposed a class of graphs, called a general biswapped graphs as following: Let G be any undirected graph with the vertex set $V(G) = \{g_1, g_2, \dots, g_n\}$ and the edge set $E(G)$. And let H be any graph with the vertex set $V(H) = \{h_1, h_2, \dots, h_m\}$ and the edge set $E(H)$. The General Biswapped Network $Gbsw(G, H)$ is a graph composed of m copies of G , $\{G_1, G_2, \dots, G_m\}$, and n copies of H , $\{H_1, H_2, \dots, H_n\}$, with additional external links in between. The vertex and edge sets of $Gbsw(G, H)$ is specified as: $V(Gbsw(G, H)) = \{g_{ij} \mid g_{ij} \in V(G_i), 1 \leq i \leq m, 1 \leq j \leq n\} \cup \{h_{ji} \mid h_{ji} \in V(H_j), 1 \leq i \leq m, 1 \leq j \leq n\}$ and $E(Gbsw(G, H)) = \{(g_{ir}, g_{is}) \mid$

$(g_r, g_s) \in E(G), 1 \leq i \leq m, 1 \leq r, s \leq n\} \cup \{(h_{jr}, h_{js}) \mid (h_r, h_s) \in E(H), 1 \leq j \leq n, 1 \leq r, s \leq m\} \cup \{(g_{ij}, h_{ji}) \mid g_{ij} \in V(G_i), h_{ji} \in V(H_j), 1 \leq i \leq m, 1 \leq j \leq n\}$.

Intuitively, the definition postulates two parts: m clusters of G (part 0) and n clusters of H (part 1) with inter-cluster links. Denote g_{ij} as $\langle 0, i, j \rangle$ and h_{ji} as $\langle 1, j, i \rangle$, representing $\langle \text{part\#}, \text{cluster\#}, \text{node\#} \rangle$, for convenience. The edge set of it is composed of three parts: intra-cluster edges in G_i (part 0), intra-cluster edges in H_j (part 1), and inter-cluster or swapping edges between two parts. From the above definition, it is clearly that G and H plays a symmetric role in the composition of $Gbsw(G, H)$. Therefore, $Gbsw(G, H)$ is isomorphic to $Gbsw(H, G)$. For example, Fig. 1 shows two specific examples of GBSNs, in which (a) is $Gbsw(K_2, C_3)$ and (b) is $Gbsw(C_3, C_4)$.

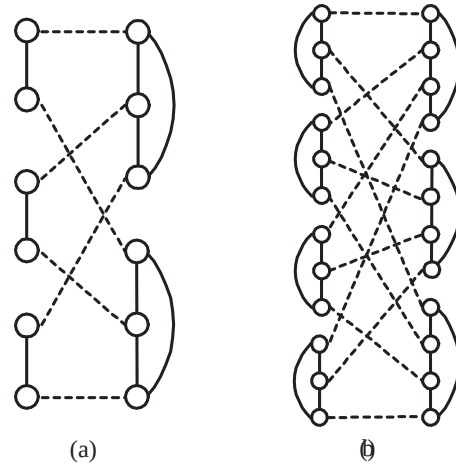


Figure 1: Examples of GBSNs. (a) $Gbsw(K_2, C_3)$ and (b) $Gbsw(C_3, C_4)$.

By definition of a general biswapped network, the following lemma is obvious obtained.

Lemma 1 $\delta(Gbsw(G, H)) = \min\{\delta(G), \delta(H)\} + 1$.

3 Diagnosabilities of general biswapped networks

Under the PMC model, a self-diagnosable system G was often modeled by a digraph in which an arc direct from vertex u to vertex v means that u can test v . In this situation, u is a tester and v is a tested vertex. We use a graph $G = (V, E)$ to represent a self-diagnosable system.

Definition 1 Under the PMC model, a syndrome σ for system G is defined as follows: For any two adjacent vertices u and v ,

$$\sigma(u, v) = \begin{cases} 0, & \text{if } v \text{ is tested by } u \text{ to be fault-free;} \\ 1, & \text{if } v \text{ is tested by } u \text{ to be faulty.} \end{cases}$$

We have the following definition of diagnosability under a precise strategy.

Definition 2 A graph G is t -diagnosable if all the faulty nodes can be identified without replacement, provided that the number of faults presented does not exceed t .

The pessimistic diagnosability measure of the diagnostic capability of a graph is defined as follows:

Definition 3 [5] A graph G is pessimistically t_1/t_1 -diagnosable if all the faulty vertices can be isolated to within a set of at most t_1 vertices, provided that the number of faulty vertices at any give time is bounded from above by t_1 . The pessimistic diagnosability of G is the maximum number t_1 such that G is pessimistically t_1/t_1 -diagnosable.

Lemma 2 [8] The graph G is t -diagnosable if and only if, for each vertex set $S \subset V(G)$ with $|S| = p$, $0 \leq p \leq t - 1$, every connected component C of $G - S$ satisfies $|V(C)| \geq 2(t - p) + 1$.

Theorem 1 For any two graphs G and H with $\delta(G) \geq 1$ and $\delta(H) \geq 1$, a general biswapped graph $Gbsw(G, H)$ is t -diagnosable where $t = \delta(Gbsw(G, H))$.

Proof. Let G and H be two graphs with $\delta(G) \geq 1$ and $\delta(H) \geq 1$ where $|V(G)| = n$ and $|V(H)| = m$. Let $t = \delta(Gbsw(G, H))$. By Lemma 1, $t \leq \delta(G) + 1$ and $t \leq \delta(H) + 1$. The argument is by contradiction.

Suppose $Gbsw(G, H)$ is not t -diagnosable. By Lemma 2, there exists a vertex set S of $Gbsw(G, H)$, $|S| = p$, $0 \leq p \leq t - 1$, such that there is a connected component C in $Gbsw(G, H) - S$, satisfying $|V(C)| \leq 2(t - p)$. Since $Gbsw(G, H)$ contains n copies of H and m copies of G , let $S_{G_i} = S \cap V(G_i)$, $S_{H_j} = S \cap V(H_j)$, $C_{G_i} = C \cap G_i$, and $C_{H_j} = C \cap H_j$. The proof is divided into two parts.

- Case 1. All vertices of C are in the same cluster, i.e., $C = C_{G_i}$ or $C = C_{H_i}$ for some i .

Without loss of generality, we may assume $C = C_{H_i}$. Since C is a subgraph of H_i ,

$\delta(C) \geq \delta(H) - r$ where $r = |N(V(C)) \cap S_{H_i}|$. Since C is a connected and $\delta(C) \geq \delta(H) - r$, $|V(C)| \geq \delta(H) - r + 1$. It is obvious that $N(V(C)) \subseteq S$. This implies $|S| \geq \delta(H) - r + 1 + |S_{H_i}| \geq \delta(H) + 1 \geq t$, which is a contradiction.

- Case 2. $C \neq C_{G_i}$ and $C \neq C_{H_j}$ for all possible i and j .

Since $2 \leq |V(C)| \leq 2(t - p)$, there is a C_{G_i} or C_{H_j} containing at most $t - p$ vertices. Without loss of generality, we may assume $1 \leq |V(C_{H_j})| \leq t - p$. Obviously, $\delta(C_{H_j}) \geq \delta(H) - r$ where $r = |N(V(C_{H_j})) \cap H_j|$ and $N(V(C_{H_j})) \cap H_j \subseteq S_{H_j}$. Hence $r \leq |S_{H_j}|$. This implies $|V(C_{H_j})| \geq \delta(H) - r + 1 \geq t - r$. Therefore, $r = p$, i.e., in this case, $S = S_{H_j} = N(V(C_{H_j})) \cap H_j$ and $|V(C_{H_j})| = t - p$. This implies $S_{G_i} = \emptyset$ and $S_{H_k} = \emptyset$ where $1 \leq i \leq m$ and $1 \leq k \neq j \leq n$. Since $\delta(G) \geq 1$ and $\delta(H) \geq 1$, C_{G_i} and C_{H_k} is connected component for $1 \leq i \leq m$ and $1 \leq k \neq j \leq n$. Therefore, $|V(C)| \geq t - p + 2(t - p) > 2(t - p)$, which is a contradiction.

Therefore, the proof is complete. $\square$

Lemma 3 [2] A graph G is a t_1/t_1 -diagnosable if and only if for any integer p , $1 \leq p \leq t_1$, and each $V' \subset V$, $|V'| = 2p$, $|N(V')| > t_1 - p + 1$.

Theorem 2 A graph G is a t_1/t_1 -diagnosable if and only if for each vertex set $S \subset V(G)$ with $|S| = p$, $0 \leq p \leq t_1 - 1$, there exists at most one isolated vertex in $G - S$ and each connected component C of $G - S$, $|V(C)| \geq 2(t_1 - p) + 1$.

Proof. ($\Rightarrow$) Suppose that there exists a $S \subset V$, $|S| = p$, $p \leq t_1 - 1$. such that $G - S$ contains at least two isolated vertices and/or there exists a component C such that $2 \leq |V(C)| \leq 2(t_1 - p)$.

Let u and v be two isolated points in $G - S$. Let $V' = \{u, v\}$. Obviously, $N(V') \subset S$. Hence $|N(V')| \leq |S| = p \leq t_1 - 1$. By Lemma 3, it is contradiction that $|V'| = 2$, $|N(V')| \geq t_1$.

Suppose that there exists a connected component C of $G - S$ that $2 \leq |V(C)| = q \leq 2(t_1 - p)$. Suppose q is an even integer. Let $k = \frac{q}{2}$. Hence $k \leq t_1 - p$. Since $0 \leq p \leq t_1 - 1$, $1 \leq k \leq t_1$. Let $V' = V(C)$. Hence $|V'| = 2k$. Since $N(V') \subseteq S$, $|N(V')| \leq |S| \leq p$. This implies $|N(V')| \leq t_1 - k < t_1 - k + 1$. By Lemma 3, it is a contradiction.

Suppose q is an odd integer. Let $k = \frac{q-1}{2}$. Hence $k < t_1 - p$. Since $0 \leq p \leq t_1 - 1$,

$1 \leq k \leq t_1 - 1$. Let $V' \subset V(C)$ such that $|V'| = q - 1$. Hence $N(V') \subseteq S \cup (V(C) - V')$ and $S \cap (V(C) - V') = \emptyset$. Hence $|V'| = 2k$ and $|N(V')| \leq p + 1 \leq t_1 - k < t_1 - k + 1$. By Lemma 3, it is a contradiction.

($\Leftarrow$) Suppose G is not a t_1/t_1 -diagnosable. By Lemma 3, there exists an integer p , $1 \leq p \leq t_1$, and $V' \subset V$, $|V'| = 2p$ such that $|N(V')| < t_1 - p + 1$.

If $p = 1$, $|V'| = 2$ and $|N(V')| < t_1$. Let $S = N(V')$. Hence $|S| = k \leq t_1 - 1$. Therefore V' is a connected component of $G - S$. Obviously V' is not an isolated vertex. Hence there must be $|V'| \geq 2(t_1 - p) + 1 \geq 3$ which is a contradiction since $|V'| = 2$.

If $2 \leq p \leq t_1$. $|V'| = 2p$ and $|N(V')| < t_1 - p + 1$. Let $S = N(V')$. Hence $|S| = k \leq t_1 - p$. This implies $0 \leq k \leq t_1 - 2$. Then there exists a connected component C in V' such that $|V(C)| \leq 2p$. Hence $|V(C)| \leq 2p \leq 2(t_1 - k)$ which is a contradiction. $\square$

Theorem 3 For any two graphs G and H with $\delta(G) \geq 1$ and $\delta(H) \geq 1$, a general biswapped graph $Gbsw(G, H)$ is t_1/t_1 -diagnosable where $t_1 = \min_{u,v \in V} \{|N(\{u, v\})|\}$.

Proof. Let $V = V(Gbsw(G, H))$ and $t_1 = \min_{u,v \in V} \{|N(\{u, v\})|\}$. Suppose $Gbsw(G, H)$ is not a t_1/t_1 -diagnosable. By Theorem 2, there exists a vertex set $S \subset V$, $|S| = p$, $0 \leq p \leq t_1 - 1$, such that (1) there are at least two isolated nodes in $Gbsw(G, H) - S$ and/or (2) there exists a connected component C in $Gbsw(G, H) - S$ satisfying $2 \leq V(C) \leq 2(t_1 - p)$.

Suppose u and v are two isolated nodes in $Gbsw(G, H)$. Obviously, $N(\{u, v\}) \subseteq S$. Hence $|S| \geq |N(\{u, v\})| \geq t_1$, which is a contradiction.

Suppose there exists a connected component C in $Gbsw(G, H) - S$ satisfying $2 \leq V(C) \leq 2(t_1 - p)$. Since $Gbsw(G, H)$ contains n copies of G and m copies of H , let $S_{G_i} = S \cap V(G_i)$, $S_{H_j} = S \cap V(H_j)$, $C_{G_i} = C \cap G_i$, and $C_{H_j} = C \cap H_j$. The proof is divided into two parts.

- Case 1. $|V(C_{G_i})| \geq 2$ or $|V(C_{H_j})| \geq 2$ for some i or some j .

Without loss of generality, we may assume $|V(C_{H_j})| \geq 2$ for some j . Let $u, v \in V(C_{H_j})$ and $r = |N(\{u, v\}) \cap S_{H_j}|$. Obviously, $r \leq |S| = p$. Since $|N(\{u, v\})| \geq t_1$, $|N(\{u, v\}) \cap V(C_{H_j})| \geq t_1 - r$. This implies $|V(C_{H_j})| \geq t_1 - r + 2$. Let $|N(V(C_{H_j})) \cap (S - S_{H_j})| = g$. Hence $g + r \leq p$. Therefore, $|V(C)| \geq 2(t_1 -$

$r + 2) - g$. This implies $|V(C)| \geq 2t_1 + 4 - p - r \geq 2t_1 - 2p + 4 > 2(t_1 - p)$, which is a contradiction.

- Case 2. $|V(C_{G_i})| \leq 1$ or $|V(C_{H_j})| \leq 1$ for all possible i and j .

Since each node $u = h_{ji}$ in cluster H_j is adjacent to only one node $v = g_{ij}$ in cluster G_i , $|V(C)| = 2$ in this case, i.e., component C is an edge (u, v) . Since $N(\{u, v\}) \subseteq S$, $|S| \geq |N(\{u, v\})| \geq t_1$, which is a contradiction.

Therefore, the proof is complete. $\square$

4 Conclusion

Some properties of general biswapped networks were studied in [4]. In this paper, we focus on the diagnosabilities of general biswapped networks under the precise and pessimistic strategies based on the PMC diagnostic model. It is proved that the general biswapped graphs $Gbsw(G, H)$ with $\delta(G) \geq 1$ and $\delta(H) \geq 1$ is $\delta(Gbsw(G, H))$ -diagnosable and t_1/t_1 -diagnosable under the precise and the pessimistic strategies, respectively, where $t_1 = \min_{u,v \in V} \{|N(\{u, v\})|\}$.

References

- [1] G. Y. Chang, G. J. Chang, and G. H. Chen, "Diagnosabilities of regular networks", *IEEE Trans. Parallel and Distributed Systems*, vol. 16, no. 4, pp. 314-323, 2005.
- [2] K. Y. Chwa and S. L. Hakimi, "On fault identification in diagnosable systems," *IEEE Trans. Computers*, vol. C-30, pp. 414-422, 1981.
- [3] J. Fan, "Diagnosability of the Möbius cubes", *IEEE Trans. Parallel and Distributed Systems*, vol. 40, no. 1, pp. 88-93, 1991.
- [4] M. He, W. Xiao, W. Chen, W. Wei and Z. Zhang, "General Biswapped Networks and Their Topological Properties", *Proc. of Advanced Parallel Processing Technologies (LNCS)*, vol. 4847, pp. 414-422, 2007.
- [5] A. Kavianpour and A. D. Friedman, "Efficient design of easily diagnosable system," in: *Proceedings of IEEE Computer Society's 3rd USA-Japan Computer Conference*, 1978.

- [6] A. Kavianpour and K. H. Kim, "Diagnosabilities of hypercubes under the pessimistic one-step diagnosis strategy," *IEEE Trans. Computers*, vol. 40, no. 2, pp. 232-237, 1991.
- [7] A. Kavianpour, "Sequential diagnosability of star graphs," *Computers Electrical Eng.*, vol. 22, no. 1, pp. 37-44, 1996.
- [8] P. L. Lai, J. J. M. Tan, C. P. Chang, and L. H. Hsu, "Conditional Diagnosability Measures for Large Multiprocessor Systems," *IEEE Transactions on Computers*, vol. 54, no. 2, pp. 165-175, 2005.
- [9] F. T. Leighton, *Introduction to Parallel Algorithms and Architecture: Arrays, Trees, Hypercubes*, Morgan Kaufmann, San Mateo, 1992.
- [10] G. Marsden, P. Marchand, P. Harvey and S. Esener, "Optical transpose interconnection system architectures," *Optics Letters*, vol. 18, no. 13, pp. 1083-1085, 1993.
- [11] F. P. Preparata, G. Metze, and R. T. Chien, "On the connection assignment problem of diagnosis systems", *IEEE Trans. Electronic Computers*, vol. 16, no. 12, pp. 848-854, 1967.
- [12] D. Wang, "Diagnosability of enhanced hypercubes," *IEEE Trans. Computers*, vol. 43, no. 9, pp. 1054-1061, 1994.
- [13] C. H. Yeh and B. Parhami, "Swapped networks: unifying the architectures and algorithms of a wide class of hierarchical parallel processors," *Proc. of Int. Conf. on Parallel and Distributed Systems, IEEE Computer Society*, pp. 230-237, 1996.
- [14] C. H. Yeh and B. Parhami, "Hierarchical swapped networks: efficient low-degree alternatives to hypercube and generalized hypercube," *Proc. Int. Symp. on Parallel Architectures, Algorithms and Networks, IEEE Computer Society*, pp. 90-96, 1996.
- [15] C. H. Yeh and B. Parhami, "Recursive hierarchical swapped networks: versatile interconnection architectures for highly parallel systems," *Proc. 8th IEEE Symp. on Parallel and Distributed Processing, IEEE Computer Society*, pp. 453-460, 1996.